HIGH-YIELD FLASHCARD DECK: IRON METABOLISM & OVERLOAD

Active Recall Preparation for Theory & Objective (MCQ) Exams

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HOW TO USE THIS DECK:

- Test yourself on each question before revealing the hidden answer.
- Pay special attention to the **High-Yield Exam Pearl** on each card.

CARD 1: Total Body Iron Distribution

- **Question:** What is the total body iron content in a healthy 70-kg adult male, and what percentage is contained within hemoglobin versus storage depots?
- **Answer:** Total body iron is **3,000 – 4,000 mg (3–4 g)**. Approximately **67% (~2,000–2,500 mg)** is in **Hemoglobin**, and **27% (~1,000 mg)** is in **Storage (Ferritin and Hemosiderin)**.
- ⚡ **Exam Pearl:** Only **0.08% (~3 mg)** is bound to transferrin in the plasma at any moment, yet this transit pool turns over 6–8 times daily to supply ~25 mg of iron to the bone marrow.

CARD 2: Daily Iron Absorption and Loss

- **Question:** What is the normal daily obligatory loss of iron from the adult body, and through what mechanism does this occur?
- **Answer: 1 to 2 mg per day**, lost through the physiological desquamation of intestinal mucosal enterocytes, epidermal shedding, and minor bile/sweat losses.
- ⚡ **Exam Pearl:** The human body possesses **no physiological excretory pathway for iron**. Therefore, iron balance is maintained exclusively at the level of intestinal absorption.

CARD 3: Heme vs Non-Heme Dietary Iron

- **Question:** Contrast the dietary sources, absorption mechanisms, and bioavailability of Heme versus Non-Heme iron.
- **Answer:**
  - **Heme Iron (ext{Fe}^{2+}):** Derived from meat, poultry, and fish. Highly bioavailable (**20–30%**), absorbed intact via putative HCP-1 transporter; unaffected by dietary chelators.
  - **Non-Heme Iron (ext{Fe}^{3+}):** Derived from plant foods, grains, and legumes. Poor bioavailability (**1–10%**); requires reduction to ext{Fe}^{2+} by apical DCYTB; enhanced by Vitamin C and inhibited by phytates, tannins, and calcium.
- ⚡ **Exam Pearl:** Ascorbic acid enhances non-heme iron absorption by maintaining iron in the soluble reduced ferrous (ext{Fe}^{2+}) form.

CARD 4: Duodenal Iron Transporters & Enzymes

- **Question:** Name the specific enzymes and transporters responsible for: (1) Apical ferric reduction, (2) Apical entry, (3) Basolateral export, and (4) Basolateral oxidation in duodenal enterocytes.
- **Answer:**
  1. Apical ferric reduction: **DCYTB (Duodenal Cytochrome b)**
  2. Apical entry: **DMT1 (Divalent Metal Transporter 1 / Nramp2)**
  3. Basolateral export: **Ferroportin-1 (Fpn / SLC40A1)**
  4. Basolateral oxidation: **Hephaestin** (membrane-bound ferroxidase)
- ⚡ **Exam Pearl:** Systemic oxidation of iron exported by macrophages into plasma is mediated by circulating **Ceruloplasmin**.

CARD 5: Hepcidin: Origin, Structure & Target

- **Question:** What is Hepcidin, where is it synthesized, what is its primary molecular target, and what is its mechanism of action?
- **Answer:** Hepcidin is a **25-amino acid peptide hormone** synthesized by **hepatocytes**. Its sole target is the iron exporter **Ferroportin**. Hepcidin binds ferroportin, inducing its **internalization and lysosomal degradation**, thereby blocking iron egress from enterocytes and macrophages.
- ⚡ **Exam Pearl:** High hepcidin traps iron inside cells, lowering serum iron (**hypoferremia**). Low hepcidin allows ferroportin to remain active, promoting maximal iron export into circulation.

CARD 6: Inflammation, IL-6 & Hepcidin

- **Question:** How does systemic inflammation cause low serum iron despite elevated bone marrow storage iron?
- **Answer:** Inflammatory cytokines (specifically **Interleukin-6 / IL-6**) activate the **JAK-STAT3 pathway** in hepatocytes, inducing **hypersecretion of hepcidin**. Hepcidin degrades ferroportin on reticuloendothelial macrophages and enterocytes, trapping recycled iron in macrophages and blocking gut absorption, resulting in hypoferremia (Anemia of Chronic Disease).
- ⚡ **Exam Pearl:** In Anemia of Chronic Disease, serum ferritin is high (acute phase reactant) while serum iron and TIBC are low.

CARD 7: Response to Hypoxia, Low Iron & Erythroferrone

- **Question:** What suppresses hepcidin synthesis when the body experiences hypoxia or acute blood loss?
- **Answer:** Hypoxia stabilizes **HIF-1alpha / HIF-2alpha**, suppressing hepcidin transcription. Concurrently, erythropoietin drives erythroblasts to secrete **Erythroferrone (ERFE)**, a potent direct suppressor of hepatic hepcidin synthesis.
- ⚡ **Exam Pearl:** Suppressed hepcidin stabilizes ferroportin, facilitating rapid mobilization of iron from macrophages and gut to support accelerated hemoglobin synthesis.

CARD 8: The Fenton Reaction

- **Question:** Write the chemical equation for the Fenton reaction and state its biological significance in iron overload.
- **Answer:** $mathbf{ ext{Fe}^{2+} + ext{H}_2 ext{O}_2 longrightarrow ext{Fe}^{3+} + ext{OH}^□ullet + ext{OH}^-} Ferrous iron catalytically decomposes hydrogen peroxide into the **hydroxyl free radical (ext{OH}^□ullet)**, which drives severe lipid peroxidation, lysosomal rupture, and mitochondrial apoptosis.
- ⚡ **Exam Pearl:** Unbound iron (Non-Transferrin-Bound Iron / Labile Plasma Iron) is the primary driver of the Fenton reaction.

CARD 9: The Haber-Weiss Reaction

- **Question:** Write the net equation for the iron-catalyzed Haber-Weiss reaction.
- **Answer:** mathbf{ ext{O}_2^{□ullet-} + ext{H}_2 ext{O}_2 longrightarrow ext{O}_2 + ext{OH}^□ullet + ext{OH}^-} Superoxide radical reduces ext{Fe}^{3+} back to ext{Fe}^{2+}, enabling catalytic amounts of iron to sustain continuous generation of hydroxyl radicals.
- ⚡ **Exam Pearl:** This explains why iron-overloaded cells suffer relentless, self-propagating oxidative damage.

CARD 10: Classical Hereditary Hemochromatosis (HFE)

- **Question:** What is the primary genetic mutation in Type 1 Hereditary Hemochromatosis, and what is the underlying molecular defect in hepcidin regulation?
- **Answer:** The **C282Y mutation in the HFE gene** on chromosome 6p (homozygous in 85–90% of cases). Defective HFE disrupts the hepatocyte iron-sensing complex (TfR2/HJV/BMP6), leading to **inappropriately low hepcidin synthesis** despite severe body iron overload.
- ⚡ **Exam Pearl:** Low hepcidin causes lifelong hyperabsorption of dietary iron (3–5 ext{mg/day}), accumulating > 20 ext{g} of iron by middle age.

CARD 11: "Bronze Diabetes" Tetrad

- **Question:** What are the components of the classic clinical tetrad in advanced, untreated hemochromatosis?
- **Answer:**
  1. **Liver Cirrhosis** (pm Hepatocellular Carcinoma)
  2. **Diabetes Mellitus** (pancreatic islet cell destruction)
  3. **Hyperpigmentation of the Skin ("Bronze Skin")** (melanin + hemosiderin)
  4. **Cardiomyopathy** (congestive heart failure and arrhythmias)
- ⚡ **Exam Pearl:** Hypogonadotrophic hypogonadism (loss of libido, testicular atrophy) and 2nd/3rd MCP arthropathy (calcium pyrophosphate pseudogout) are also prominent.

CARD 12: Ineffective Erythropoiesis & Iron Overload

- **Question:** Why do patients with non-transfused Beta Thalassemia Intermedia develop severe iron overload?
- **Answer:** Massive intramedullary apoptosis of erythroid precursors leads to massive **Erythroferrone (ERFE)** release, which **completely suppresses hepcidin**, causing unregulated intestinal iron hyperabsorption even without transfusions.
- ⚡ **Exam Pearl:** Iron overload can result from excessive intake, repeated transfusions, or ineffective erythropoiesis.

CARD 13: Laboratory Differentiation: IDA vs ACD

- **Question:** Contrast the laboratory values of Ferritin, TIBC, and Soluble Transferrin Receptor (sTfR) in Iron Deficiency Anemia versus Anemia of Chronic Disease.
- **Answer:**
  - **IDA:** Ferritin is **Low (< 15 mu ext{g/L})**; TIBC is **High (> 400 mu ext{g/dL})**; sTfR is **Elevated**.
  - **ACD:** Ferritin is **Normal or High (> 100 mu ext{g/L})**; TIBC is **Low / Normal (< 250 mu ext{g/dL}$)**; sTfR is **Normal**.
- ⚡ **Exam Pearl:** Serum Ferritin is the single best blood test to differentiate uncomplicated IDA from ACD.

CARD 14: Treatment: Phlebotomy vs Chelation

- **Question:** When is therapeutic phlebotomy indicated for iron overload, and when is it contraindicated? Name three iron chelating drugs.
- **Answer:**
  - **Phlebotomy:** First-line for **Hereditary Hemochromatosis** (where hemoglobin is normal).
  - **Contraindicated in: Secondary (Transfusional) Iron Overload** because the patient has severe baseline anemia (e.g., Thalassemia Major).
  - **Iron Chelators:**
    1. **Deferoxamine (DFO):** Subcutaneous/IV infusion over 8–12 hours.
    2. **Deferasirox (DFX):** Once-daily oral tablet/suspension.
    3. **Deferiprone (DFP):** Oral tablet (causes agranulocytosis; optimal for cardiac iron).
- ⚡ **Exam Pearl:** Weekly phlebotomy removes ~200–250 mg of elemental iron per 500 mL of whole blood.